

# The Fable Computer, Part II: Quantum-Limited Analog Tensor Processing on the Regenerative Graphene-Plasmon Fabric

*Interference computing with mesoscopic plasmon pulses, classical on-chip decoding to 1–2 bits, and the temperature budget of the quantum–classical crossover*

Ryoji Furui

www.ryoji.info

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*Authorship and Status Disclaimer. This manuscript extends the author's design study [1] ("Part I" throughout). Generation of content involved the use of an AI, Anthropic's Fable 5, at the author's direction, including construction and adversarial cross-checking of the released model package (Appendix QA). The author declares a lack of formal expertise in quantum optics and graphene-plasmonic device physics, and the document has not received independent expert review. All claims are reduced-order model results in the sense of Part I ("in-model": established within the released chain, under its stated assumptions, not measured), and must be treated as a starting point for discussion, not as peer-reviewed engineering guidance.*

## Abstract

Part I designed a room-temperature terahertz half adder on a regenerative graphene-plasmon fabric and answered its feasibility question in-model. This Part asks what the same fabric computes when its pulses are treated as what they are: quantum fields. Quantizing the Part-I Dyakonov–Shur (DS) cell mode puts one plasmon at a fractional density amplitude of 0.16 %, the 1-dB compression knee at  $\approx 38$  plasmons and the logic rail at  $\approx 3.8 \times 10^3$  — logic pulses are mesoscopic, and quantum noise is a quantitative design input, not a curiosity. That ladder is mode-energy quantization applied to two classical Part-I solver amplitudes; it fixes the mesoscopic framing and sizes the machine, and — as the merged community record establishes — it is not what the no-go below rests on. Two results follow. First, an honest no-go: the single-plasmon Kerr nonlinearity of the 576-nm cell falls short of its loss rate by roughly five orders of magnitude ( $\chi/\kappa \approx 8.2 \times 10^{-6}$  at 4 K at the carried Kerr coefficient;  $\approx 4.5$ – $5.8$  orders at 4 K across its open  $c_K = 0.05$ – $1$  band), so gate-model (photon-blockade) quantum logic is excluded on this fabric at any temperature; the blockade would demand  $\sim 3$ -nm cells, consistent with the quantum-plasmonics literature. Second, what survives is exactly the machine the fabric's own architecture suggests: a quantum-limited analog (continuous-variable) computer, in which comb-locked coherent pulses interfere at gate-programmed junctions — plasmon–plasmon interaction in the analog-computer sense — and the result is quantized on-chip by the Part-I threshold cells themselves, re-programmed as a 1–2-bit flash decoder. The minimal tensor unit, QMAC-1 (two inputs, one interference, one output decoded to 2 bits), is specified to cell-map level; in its digital special case the decoded word is the binary count  $A+B$ , so the Part-I half adder re-emerges as the decoder of a single quantum interference. Error rates versus temperature are computed on one axis for the quantum-analog unit and the classical digital fabric: the 2-bit decode is unusable per-shot at 300 K (0.14), reaches  $3.1 \times 10^{-3}$  at 77 K, and saturates at a vacuum-set floor of  $1.3 \times 10^{-6}$  below  $\approx 20$  K ( $N = 400$  plasmons per input); the 1-bit decode is already usable warm ( $3.9 \times 10^{-3}$  at 300 K) — lower-bit decoding is what the physics supports per shot.  $T_Q = \hbar\omega/k_B \approx 48$  K is where the noise budget crosses from thermal- to vacuum-

dominated; below  $\approx 15\text{--}20$  K the error curves saturate at vacuum-set floors that no further cooling improves. That saturation is the quantum limit made visible, and its observation is a pre-registered bench gate. Part I's no-cryogenics lock is explicitly relaxed here and the fork is stated as such.

## 1. Introduction: from restored bits to quantum-limited amplitudes

Part I [1] built a machine out of one physical object — a resonant DS cell [2] biased below its self-oscillation threshold — and one discipline: every gate restores its logic levels against a rail, so decisions never erode. The price of that discipline is nonlinearity: digital restoration lives at the saturating rail. This Part starts from the opposite end of the same transfer curve. Below the 1-dB knee the fabric is a linear, phase-coherent wave medium: pulses split, merge, and interfere at gate-programmed junctions, and the comb-locked clock of Part I — distributed at femtosecond-class phase — gives every cell a common phase reference. A linear interferometer with programmable weights is an analog computer: it multiplies by transmission amplitudes and adds by superposition. The question Part II asks is how well that analog computer can compute, and the answer is a quantum one, because the pulses in play are small.

Quantizing the Part-I operating point (Section 3) puts the knee of a cell at  $\approx 38$  plasmon quanta and the rail at  $\approx 3833$  — mode-energy bookkeeping applied to two classical solver amplitudes, which is what makes the framing mesoscopic and what sizes the machine, but which is not, and is not used as, an argument about quantum logic (Section 3.3). At those occupations three quantum facts dominate the machine design. The 1-THz quantum is small against the room-temperature bath ( $\hbar\omega/k_B \approx 48$  K), so at 300 K every mode carries  $\approx 5.8$  thermal plasmons of noise; below  $\approx 48$  K vacuum fluctuations take over — the exact crossover, where the thermal occupation falls to the vacuum half-quantum, is  $T^* = \hbar\omega_0/(k_B \ln 3) \approx 43.7$  K (promoted community derivation, July 2026) — and by  $\approx 15\text{--}20$  K the noise floor stops improving altogether. Phase-insensitive gain — the only kind the DS cell provides [2, 3] — adds at least the Caves quantum noise, which is precisely the  $F = 2 - 1/G$  floor Part I already carried as its Eq. (7). And amplitude resolution per shot is bounded by shot noise, so the number of bits one pulse can carry is finite and computable. The machine that emerges is not a quantum computer in the gate-model sense — Section 3.3 quantifies why that is excluded on this cell — but a quantum-limited analog tensor engine with a classical digital wrapper: the same fabric, the same five-cell block as Part I, re-programmed so that two of its cells act as comparators of a flash analog-to-digital ladder.

One constraint of Part I is explicitly relaxed: no-cryogenics. The quantum-analog unit works at 300 K only in its 1-bit or averaged modes; its 2-bit per-shot mode wants  $\leq 77$  K, and its quantum floor lives below  $\approx 20$  K (Section 6). Part II therefore defines temperature classes rather than one operating point, keeps the  $\leq 80$  °C cap for the warm class, and states plainly that the cold classes are a fork from Part I's charter, not a refinement of it. Cooling also changes the classical fabric itself (Part I, Section 8.3): the DS threshold tracks temperature, so all cold-side operation inherits Part I's tracked-bias requirement, and below  $\approx 250$  K the hydrodynamic tier that certifies the gain numbers expires — every cold-side gain figure here carries that caveat to the Boltzmann–Maxwell tier, exactly as in Part I.

## 2. Related work and positioning

Quantum plasmonics. That surface plasmons preserve quantum coherence is established: entangled photons survive conversion through plasmonic channels [4], and the field has a developed single-quantum toolbox [5]. For graphene specifically, the extreme mode confinement prompted the prediction that few-nanometre graphene nanostructures could reach single-plasmon Kerr nonlinearities of order unity [6] — the photon-blockade regime. Section 3.3 places the Part-I cell against exactly that scaling and finds it roughly five orders of magnitude away ( $\approx 4.5$ – $6.1$  orders across the open c\_K band at the two temperatures Section 3.3 quotes): this design claims the opposite corner of quantum plasmonics, many-quantum Gaussian operation at the standard quantum limit.

Continuous-variable computing. Encoding analog values in field quadratures and processing them with Gaussian operations is the continuous-variable framework [7, 8]; universal quantum computation in that framework is known to require a non-Gaussian resource [7], which is precisely what the Kerr no-go denies this fabric. What Gaussian operations do provide is quantum-limited analog arithmetic: the amplifier bound of Caves [9] (anticipated by Haus and Mullen [10]) sets the noise cost of gain, and homodyne-class detection against vacuum noise sets the per-shot precision.

Photonic analog accelerators. Coherent interferometric multiply-accumulate is the operating principle of photonic neural accelerators, from programmable nanophotonic processors [11] to comb-based convolutional tensor cores [12]. The niche claimed here is different in three ways: the carrier is a THz plasmon confined  $\approx 100\times$  below the free-space wavelength (footprint), the weights and routing are gate-voltage-programmable on one chip, and — the load-bearing difference — the analog result is quantized by the same regenerative cell family that computes, so no electronic ADC enters the signal path (Section 5). The cost is the THz quantum: at 1 THz the thermal bath is hot in quantum units at room temperature, which is why Section 6's temperature budget is the central result of this Part.

## 3. Quantum scales of the Part-I operating point

### 3.1 How many quanta is a logic pulse?

The hydrodynamic energy of the acoustic (gate-screened) standing mode [13] of one DS cell (length  $L \approx 576$  nm, effective area  $A \approx 10^{-8}$  cm<sup>2</sup>, density  $n = 10^{12}$  cm<sup>-2</sup>, speed  $s$  from Part-I Eq. (1)) with fractional density amplitude  $\varepsilon = \delta n/n$  is

$$E(\varepsilon) = \frac{1}{4} n m^* s^2 \varepsilon^2 A, \quad m^* = E_F / v_F^2 \quad (\text{Q1})$$

(kinetic plus potential of the shallow-water mode, spatially averaged over the standing wave). Setting  $E = \hbar\omega_0$  defines the single-plasmon amplitude and the quantum content of any classical swing:

$$N(\varepsilon) = (\varepsilon/\varepsilon_1)^2, \quad \varepsilon_1 = \sqrt{(4\hbar\omega_0 / n m^* s^2 A)} \approx 1.62 \times 10^{-3} \quad (\text{Q2})$$

One plasmon is a 0.16 % density swing. The Part-I solver's 1-dB gain-compression knee ( $\varepsilon \approx 1$  %) is therefore  $N \approx 38$  quanta (0.025 aJ intracavity), and the  $\approx 10$  % rail is  $N \approx 3833$  (2.5 aJ). The provenance of these two numbers should be stated exactly (promoted community derivation, July 2026): both  $\varepsilon$  values are classical Part-I shallow-water-solver outputs — the

compression knee and the gain-saturation rail — imported here and converted to quanta by Eqs. (Q1–Q2). They are quantization bookkeeping applied to classical inputs, not outputs of the quantum noise chain, which is exactly linear and amplitude-scale-free in  $N$  and can generate no intrinsic occupation threshold of its own (Section 4.1). The arithmetic reproduces from Eqs. (Q1–Q2) against an independent hand reconstruction to  $\approx 1.2\%$  in  $\varepsilon_1$  and, because  $N \propto \varepsilon_1^{-2}$ , to  $\approx 2.4\%$  in  $N_{\text{knee}}$  and  $N_{\text{rail}}$  (Appendix QA); the absolute calibration of Eq. (Q1) — the effective area  $A$  and the  $\frac{1}{4}$  mode-energy prefactor, one hydrodynamic averaging convention — is owned by the Boltzmann–Maxwell tier (Section 8) and is open: a factor of two in the prefactor moves both  $N_{\text{knee}}$  and  $N_{\text{rail}}$  by two. Two scales should not be confused. Part I's “intrinsic field energy of  $\approx 10$  aJ per bit” counts a traveling few-cycle pulse over its  $\approx 3\lambda_p$  in-flight extent (an area an order of magnitude larger than one cell); Eqs. (Q1–Q2) count quanta in the quarter-wave cavity mode of one cell, which is the mode the noise physics and the comparator decisions live in. The two are consistent — a traveling pulse at rail-class swing over  $3\lambda_p$  carries 10–30 aJ — and both sit below the launch-side slot energy before the still-unquantified coupling (Part I, Section 9.5). A logic pulse on this fabric is a mesoscopic object — tens to thousands of quanta — and Figure Q1(A) shows the whole ladder. What that ladder anchors is the mesoscopic framing of this Part and the classical machine design that follows from it — the passive-core rule of Section 4.2 and the decoder sizing of Section 5. It does not anchor the no-go of Section 3.3, whose logic runs on entirely different quantities; the two roles are kept separate throughout.

### 3.2 The bath: 48 K in quantum units

The 1-THz quantum is  $\hbar\omega_0 \approx 4.14$  meV, so the crossover temperature is  $T_Q = \hbar\omega_0/k_B \approx 48$  K. Table Q1 gives the Bose–Einstein occupation of the mode across the temperatures used in this Part. At the Part-I band (300–353 K) every mode carries six to seven thermal quanta — the fabric computes while swimming in a bath an order of magnitude above the vacuum floor; at 77 K one thermal quantum remains; at 20 K the mode is effectively dark. The plasmon lifetime improves in parallel but saturates:  $\tau(T)$  follows Part I's  $1/T$  phonon scaling down to  $\approx 150$  K and is clamped by impurity scattering below (Part I, Section 8.3), so  $Q$  saturates at  $\approx 12.6$ . The saturated  $\tau = 2$  ps sits slightly above the best measured cryogenic graphene-plasmon lifetimes ( $\approx 1.6$  ps [14]) — a premium-material assumption in exactly the sense of Part I's  $\tau(300\text{ K}) = 1$  ps, and flagged as such. Cooling buys a quieter bath much faster than it buys a better resonator.

| T (K)         | $\bar{n}$ (thermal quanta) | $Q = \omega\tau$ | per-gate loss (dB, $\lambda_p/2$ ) |
|---------------|----------------------------|------------------|------------------------------------|
| 353           | 6.87                       | 5.3              | 2.56                               |
| 300           | 5.76                       | 6.3              | 2.17                               |
| 150           | 2.65                       | 12.6             | 1.09                               |
| 77            | 1.16                       | 12.6 (sat.)      | 1.09                               |
| 48 (= $T_Q$ ) | 0.58                       | 12.6 (sat.)      | 1.09                               |
| 20            | 0.10                       | 12.6 (sat.)      | 1.09                               |
| 4             | $6.2 \times 10^{-6}$       | 12.6 (sat.)      | 1.09                               |

Table Q1. The 1-THz mode against its bath. Lifetime saturation below  $\approx 150$  K follows Part I's stated cold-side caveat; premium (graphite-gated) material is carried as upside, not assumed.

### 3.3 The honest no-go: no photon blockade on this cell

Gate-model quantum logic needs one quantum to shift the cell's resonance by more than a linewidth —  $\chi/\kappa \gtrsim 1$ , with  $\chi$  the single-plasmon Kerr rate and  $\kappa = 1/\tau$  the decay rate. The hydrodynamic nonlinearity gives, at reduced order,

$$\chi \approx c_K \omega_0 \varepsilon_1^2, \quad c_K = O(1) \text{ band } 0.05\text{--}1 \text{ (0.25 carried)} \quad (\text{Q3})$$

i.e.  $\chi/2\pi \approx 0.7$  MHz against  $\kappa/2\pi \approx 160$  GHz at 300 K:  $\chi/\kappa \approx 4.1 \times 10^{-6}$  at 300 K and  $8.2 \times 10^{-6}$  even at the lifetime-saturated 4 K — five orders of magnitude short at the carried  $c_K = 0.25$ , and  $\approx 4.5\text{--}6.1$  orders once the open  $c_K = 0.05\text{--}1$  band of Eq. (Q3) is taken in at both of the temperatures quoted here ( $\approx 4.5\text{--}5.8$  orders at 4 K,  $\approx 4.8\text{--}6.1$  at 300 K) — short at any temperature this stack can reach, and nowhere by less than  $\approx 4.5$  orders, since the lifetime-saturated cold rows are the most favourable case the stack offers and warming only shortens  $\tau$  and widens the gap. That band is the honest width of the “five orders” phrase (promoted community derivation, July 2026): the shortfall is robust at order-of-magnitude level, but its exponent is only as sharp as  $c_K$ , which the Boltzmann–Maxwell tier owns (Section 8). Because  $\chi \propto 1/A$  at fixed density and frequency, closing the gap would demand  $\approx 3$  nm structures — a bookkeeping scale, not a device: no few-nanometre cell hosts a 1-THz quarter-wave mode at all (its resonance would sit near 200 THz), so blockade is excluded here twice over, by magnitude and by geometry. Both exclusions are consistent with the graphene quantum-plasmonics literature, whose blockade proposals live at few-nanometre scale and correspondingly mid-infrared frequencies [6]. The conclusion is stated plainly: this fabric supports no plasmonic qubit. Its load-bearing anchors are named as plainly: in-model the exclusion rests on the  $\chi/\kappa$  shortfall, with its open  $c_K$  band, and on the geometry. The 38/3833 ladder of Section 3.1 is not among them — that ladder is quantization bookkeeping on classical solver amplitudes and plays no part in this exclusion (promoted community derivation, July 2026). Nor is the no-go fragile for resting on two things; it is over-determined. The released chain's own verified loss+thermal generator dephases a superposition of vacuum and an N-quantum pulse within a single gate segment for N of order unity at every temperature in this Part ( $N^* \approx 0.4$  at 300 K,  $\approx 9$  at 4 K, per gate), which excludes a plasmonic qubit here more brutally than the Kerr gap does. That decoherence factor is a standard consequence of the Lindblad generator verified in Appendix QA and not a printed output of the released code; it is cited as a hand derivation, and its proximity to the knee at 4 K is a coincidence of two quantities with no shared parameter dependence, not a corroboration. The mesoscopic cross-Kerr interaction is real but sub-resolution per shot — a 400-quantum control pulse writes  $\approx 6.6$  mrad of phase in 2 ps, versus  $\approx 25$  mrad single-shot homodyne resolution at 4 K ( $\approx 15$  shots to resolve): measurable as a systematic with averaging (bench gate QG5), not computational per slot. Everything that follows is therefore Gaussian: coherent states, linear optics, phase-insensitive gain, and homodyne-class decisions — quantum-limited analog computing, with the limits computed rather than assumed.

Figure Q1. Quantum scales of the plasmon fabric (operating point of Part I)

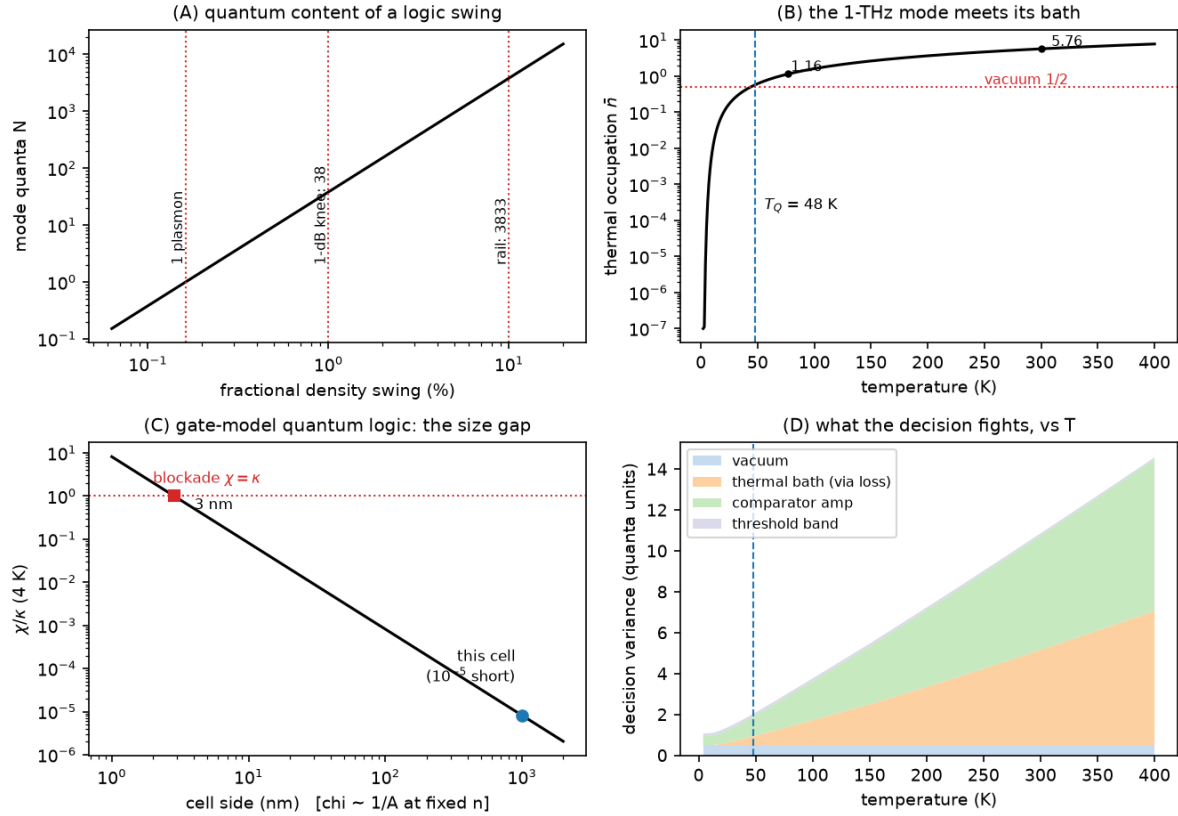


Figure Q1. Quantum scales of the fabric at the Part-I operating point. (A) Quanta in a classical swing, Eq. (Q2): one plasmon, the 38-quantum knee, the  $3.8 \times 10^3$ -quantum rail. (B) Thermal occupation of the 1-THz mode:  $T_Q \approx 48$  K separates the thermal and vacuum regimes. (C) The blockade gap:  $\chi/k$  versus cell size at fixed density; the Part-I cell sits five orders below blockade at the carried  $c_K = 0.25$  ( $\approx 4.5$ – $5.8$  orders at 4 K across its open band), which would demand  $\approx 3$ -nm cells [6]. (D) Decision-noise budget versus temperature: vacuum, thermal bath, comparator amplifier noise, and threshold-band terms (Section 5).

## 4. The computing model: quantum-limited interference

### 4.1 Gaussian rules on the fabric

Pulses are comb-locked coherent states; in the measured quadrature  $x$  (vacuum variance  $1/2$ ) a pulse of  $N$  quanta has mean  $\sqrt{(2N)}$ . Three operations exhaust the analog layer, each with an exact Gaussian rule: propagation loss  $\eta$  at temperature  $T$  mixes in the bath,  $V \rightarrow \eta V + (1-\eta)(\bar{n} + 1/2)$ ; a junction combiner with amplitude weights  $(\sqrt{w}, \sqrt{(1-w)})$  superposes means and averages variances; and regenerative gain  $G$  adds the matched-temperature phase-insensitive amplifier noise,  $V \rightarrow GV + (G-1)(\bar{n} + 1/2)$ . The last rule is Part I's noise model made explicit: referred to a matched input floor it reproduces  $F = 2 - 1/G$  (Part-I Eq. (7)) at every temperature, and reduces to the Caves bound [9] as  $T \rightarrow 0$ . (Provenance, community review July 2026: the comparator floor computed here at 300 K, 2.775 dB, and Part I's 2.77 dB floor, evaluated at 353 K, are one closed form at two temperatures — near  $G \approx +9.7$  dB the floor moves only  $\approx 0.06$  dB per dB of gain — so the numerical agreement between the two manuscripts is structurally

forced, not an independent cross-check. The promoted derivation note makes this falsifiable: exported at 353 K,  $F_{\text{dec}}$  must reproduce the parent floor to float precision, 2.768939660565078 dB; below 150 K the two repositories' floors diverge by construction — this repo's saturated  $\tau_q$  gives 2.7925 dB at 77 K against 2.8012 dB from the parent's unsaturated lifetime — a designed separation, not a discrepancy.) A further promoted derivation (July 2026) fixes the scope exactly: the noise figure carries no quantum correction at any plasmon number — the matched floor's  $\hbar$  cancels identically, at the 38-quantum knee exactly as at the rail — so the quantum/classical separation of this Part is a temperature phenomenon ( $T^* \approx 43.7$  K), never an occupation-number one. A second promoted derivation (July 2026) draws the structural corollary: because these rules are exactly linear and the noise figure carries no  $1/N$  correction, the chain is amplitude-scale-free in  $N$  and can generate no intrinsic plasmon-number threshold at all — asked its own coherence question, when single-plasmon coherence becomes classically indistinguishable, it answers  $N^* \approx 0.02$ –9 depending on temperature and criterion, one to four orders below the mesoscopic ladder of Section 3.1 and with different parameter scaling, so no tuning reconciles the two. Every occupation scale in this Part — the knee, the rail,  $N_{\text{op}}$  — is imported from the classical solver or the launch budget, never derived here; the warm band's absolute variances carry a 0.31–0.36 dB vacuum share, classical-equivalent in structure though not to the 0.1 dB level. All three rules are verified in the released package against exact Fock-space quantum evolution (truncated-basis Lindblad for loss, two-mode squeezing for gain) to better than 0.01 % (Appendix QA). One deliberate conservatism: decisions are modeled as midpoint-threshold homodyne discrimination — what the comparator cells physically implement — rather than the optimal joint quantum measurement, whose Helstrom bound [15] would improve the error exponent by at most a factor of two for these level sets.

Weights are programmable the way everything on this fabric is programmable: a junction's split ratio is a gate-map choice, and a weight's sign (phase) is a programmed delay — the plasmon speed under a segment is density-tuned through Part-I Eq. (1), so a half-wave of extra path is a voltage, not a mask change. The interference core therefore implements  $y = \sum w_i x_i$  with amplitude weights of either sign: a full four-quadrant multiply-accumulate.

## 4.2 Two computed design rules

The analog core must be passive. The regenerative cell's linear window ends at the  $\approx 38$ -quantum knee, so any pre-amplifier caps the analog signal exactly where thermal noise is most damaging. Computed head-to-head (Appendix QA): the pre-amplified variant of the tensor unit (knee-capped at  $N_{\text{op}} = 19$ ) delivers 0.50 symbol error at 300 K and 0.10 even at 4 K — against 0.14 and  $1.3 \times 10^{-6}$  for the passive core at  $N_{\text{op}} = 400$ . Phase-insensitive gain before quantization loses at every temperature on this fabric; gain belongs after the decision, inside the decoder cells, which are threshold devices and want the rail anyway. This is the quantum restatement of Part I's own discipline: restore digitally, never amplify analog. Note where the  $\approx 38$ -quantum knee is load-bearing: here, and in the decoder sizing of Section 5 — in the classical machine design, that is, and not in the no-go of Section 3.3 (promoted community derivation, July 2026). The rule therefore inherits the open absolute calibration of Eq. (Q1) along with the knee itself; the head-to-head margin is orders of magnitude wide, so a factor-of-two shift in the knee would not move the verdict, but the dependence is real and is stated rather than assumed away.

The launch must be quiet. A launch carrying Part I's budgeted 20-dB classical SNR (modeled as amplitude noise at a fixed fraction of the full swing, on both arms) pins the decode error near  $3.4 \times 10^{-3}$ – $1.9 \times 10^{-2}$  at every temperature — classical launch noise does not cool away (Figure Q3, grey curve). Quantum-limited operation requires a near-shot-noise-limited launch, which the photonic chain of Part I does not currently promise. This is Part II's sharpest new hardware requirement and is carried as an open item alongside Part I's launch-coupling efficiency.

## 5. QMAC-1: the minimal tensor unit, decoded to 2 bits

The smallest complete tensor operation is one weighted sum of two inputs, quantized. QMAC-1 (Figure Q2) is that unit on the fabric: two comb-locked input pulses ( $N_{\text{op}}$  quanta each at logic 1;  $N_{\text{op}} = 400$  default, launch-budget-capped) pass gate-programmed weight segments, interfere in one junction combiner, and the output amplitude is split to two comparator cells — Part-I threshold cells with staggered, trimmed thresholds and sharpness  $k \geq 16$  — whose thermometer code is mapped to binary by one inverting cell and one AND cell. Five active cells,  $\approx 12 \times 6$  lattice sites ( $\approx 13 \times 14 \mu\text{m}$  at the Part-I site pitch, Figure Q5), one MAC per 4-ps slot, wave-pipelined like everything on this fabric.

The decoder is not new hardware. Comparators, inverter, AND: these are the Part-I gate vocabulary, and the block is cell-for-cell the Part-I half adder with its input stage re-trimmed as a two-level flash ladder. In the digital special case (both weights  $\frac{1}{2}$ , inputs  $\in \{0, N_{\text{op}}\}$ ) the interference levels are 0,  $N_{\text{op}}/2$  and  $2N_{\text{op}}$  — equally spaced in amplitude — and the decoded word  $b_1 b_0$  is the binary count  $A+B$ : sum and carry (Table Q2). The Part-I half adder re-emerges as the decoder of a single quantum interference — the demonstrator of Part I and the decoder of Part II are the same five cells, re-programmed.

| A | B | interference level (quanta) | $t_2 t_1$ | $b_1 b_0$ | = A+B |
|---|---|-----------------------------|-----------|-----------|-------|
| 0 | 0 | 0                           | 0 0       | 0 0       | 0     |
| 0 | 1 | $N_{\text{op}}/2$           | 0 1       | 0 1       | 1     |
| 1 | 0 | $N_{\text{op}}/2$           | 0 1       | 0 1       | 1     |
| 1 | 1 | $2 N_{\text{op}}$           | 1 1       | 1 0       | 2     |

Table Q2. QMAC-1 digital truth table: the 2-bit decode of one interference is the half adder of Part I ( $b_1$  = carry,  $b_0$  = sum). Verified digitally in the released package.

In analog mode the same unit computes  $y = \sqrt{w} \cdot x_A + \sqrt{(1-w)} \cdot x_B$  and the ladder quantizes  $y$  to 2 bits. The per-shot analog precision — effective bits of one MAC against the full noise budget — is 0.6 bits at 300 K, 2.0 at  $T_Q$  and 2.5 at 4 K (Figure Q4A): the physics itself says this fabric is a 1–2-bit-per-shot analog machine, which is why the decoder is sized at 2 bits and why lower-bit decoding is not a concession but the design point. A worked weighted example ( $w = 0.7$ ,  $5 \times 5$  input grid, 77 K) holds worst-case rms noise at 0.36 LSB of the 2-bit ladder (Appendix QA). Deeper precision is bought with slots: accumulating 16 slots in the electronic charge-memory layer (Part I, Section 6.4) averages the per-slot noise (the decision is taken once on the accumulated mean, so the per-slot variance falls as  $1/16$  — mean accumulation, not a majority vote over per-slot decisions, which would scale far more weakly — a 9-of-16 majority vote is  $\approx 175 \times$  worse at 300 K) — though not the static threshold band, which is common to all slots —



and gains five decades at the Part-I band (Section 6) at 1/16th throughput, still  $1.6 \times 10^{10}$  MAC/s per unit. One design balance was recorded here in v1.3 and is corrected in v1.5, because it was stated without its reference plane (notes/2026-07-14-q2bit-avg16-averaging-convention.md §1.5, promoted 3-of-3): at the  $k = 16$  rule the static band's variance is  $(x\_range/k)^2/12$ , which evaluated on the pre-loss signal swing  $x\_range = 2\sqrt{N}$  is  $N/768 \approx 0.52$  at  $N_{op} = 400$  and therefore crosses the vacuum floor ( $\frac{1}{2}$ ) at  $N \approx 384$  — apparently the design point exactly. That crossing is real but frame-dependent, and the frame must be named. The band is a signal-derived quantity and attenuates with the swing, whereas the vacuum floor is the fixed point of the chain's loss map  $V \rightarrow \eta V + (1-\eta)(\bar{n} + \frac{1}{2})$  and does not attenuate at all. At the plane where the decision is actually taken — the comparator input, after the chain's losses —  $x\_range \approx 17.5$  (300 K) to 19.9 (4 K), so the static band is  $V_{static} \approx 0.10$ – $0.13$ , some 4–5× below the invariant  $\frac{1}{2}$ , and the band-overtakes-vacuum crossover moves up by that same factor, to  $N \approx 2 \times 10^3$  at 300 K. What survives is narrower than the claim it replaces, and it is a comparison of two terms rather than a statement about the whole decision: at the decoder plane and at  $N_{op} = 400$  the static band sits well below the vacuum half-quantum at every temperature in this Part, so sharpening comparators beyond  $k = 16$  buys little per shot. Which term then binds is a matter of temperature: only at cold temperatures, where  $\bar{n} \ll \frac{1}{2}$ , is the vacuum floor what the per-shot decision runs into; through the Part-I band the decision is thermal-limited ( $V_{slot} \approx 10.8$  at 300 K) and neither the band nor vacuum is what sets it. The coincidence of  $N \approx 384$  with  $N_{op} = 400$  is a property of the signal-referred frame, not a design fact about the comparator. In the 16-slot averaged mode the accounting differs again, because the per-slot terms average and the band does not (Section 6). No released number changes; the two figures are the same term metered one attenuation apart.

Figure Q2. The minimal quantum-analog tensor unit and its 2-bit decode ( $N_{op} = 400$  quanta)

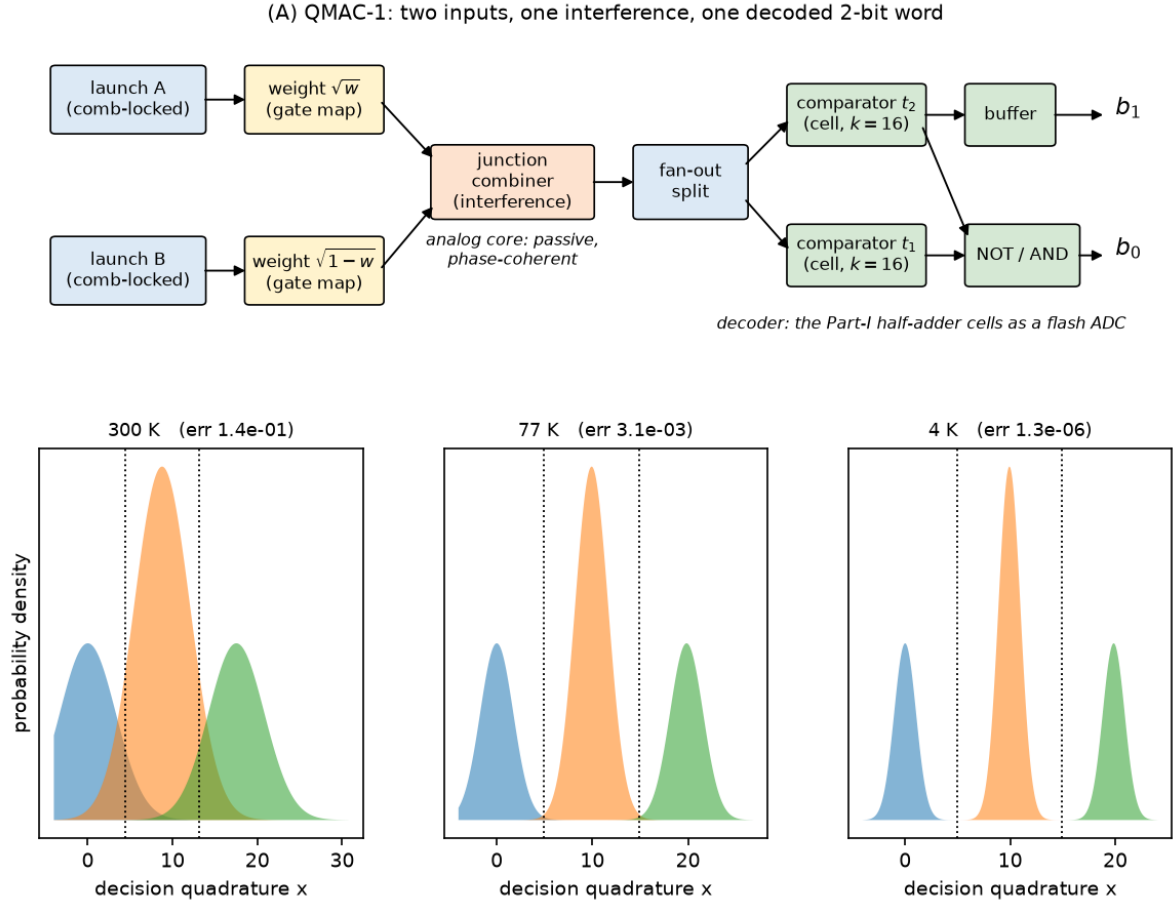


Figure Q2. QMAC-1. (A) Cell-level netlist: comb-locked launches, gate-programmed weights, one junction combiner (the interference that computes), fan-out split, two comparator cells, and the thermometer-to-binary cells; the decoder is the Part-I half-adder block re-programmed. (B) The three interference levels and their Gaussian noise at 300, 77 and 4 K with midpoint thresholds ( $N_{op} = 400$ ): the same levels drawn at 300 K, separate at 77 K, and sit an order of magnitude apart at the vacuum floor.

Figure Q5. QMAC-1 on the gate lattice:  $\sim 12 \times 6$  sites ( $\sim 13 \times 14 \mu\text{m}$  at the Part-I site pitch), five active cells - the Part-I block, re-programmed

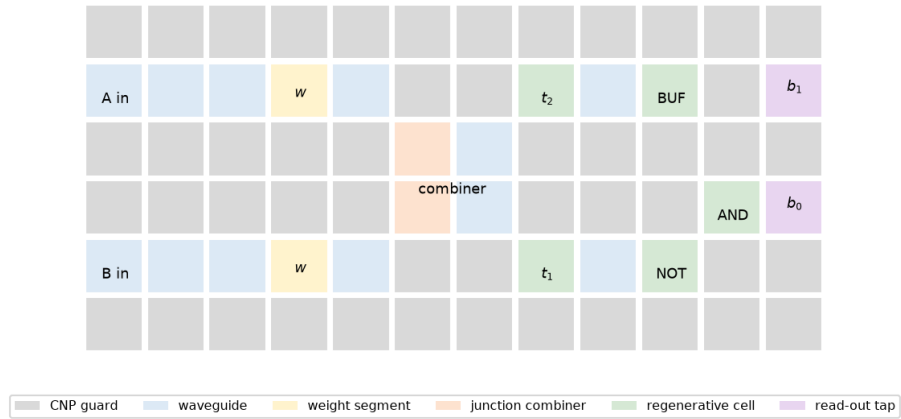


Figure Q5. QMAC-1 on the gate lattice (Part-I Figure 5b convention):  $\approx 12 \times 6$  sites including guards, five active cells, weights and combiner as programmed junction segments, read-out taps at the boundary. The map is a re-programming of the Part-I half-adder block, not a new tile.

## 6. Error rates versus temperature

Figure Q3 and Table Q3 put the quantum-analog unit and the classical digital fabric on one temperature axis, 4–400 K, holding the hardware fixed and letting only  $\bar{n}(T)$ ,  $\tau(T)$  and the loss/gain budget move. The classical-digital curve is this package's own construction (Part I never quotes a bit-error rate): a rail-restored decision — swing at the  $\approx 10\%$  rail ( $\approx 3833$  quanta), Part I's static noise margin (0.256 of swing), against the matched-temperature amplifier noise of one restoring cell. It is the physics floor of the digital fabric, with technical noise excluded on both sides of the comparison.

| T (K) | $\bar{n}$            | classical digital BER | quantum-analog 2-bit | quantum-analog 1-bit  | 2-bit, 16-slot avg    | analog bits/shot |
|-------|----------------------|-----------------------|----------------------|-----------------------|-----------------------|------------------|
| 353   | 6.87                 | $9.1 \times 10^{-10}$ | $1.8 \times 10^{-1}$ | $9.6 \times 10^{-3}$  | $6.5 \times 10^{-6}$  | 0.43             |
| 300   | 5.76                 | $3.6 \times 10^{-11}$ | $1.4 \times 10^{-1}$ | $3.9 \times 10^{-3}$  | $4.7 \times 10^{-7}$  | 0.62             |
| 150   | 2.65                 | $2.4 \times 10^{-20}$ | $2.5 \times 10^{-2}$ | $1.1 \times 10^{-5}$  | $2.2 \times 10^{-13}$ | 1.29             |
| 77    | 1.16                 | $5.7 \times 10^{-37}$ | $3.1 \times 10^{-3}$ | $4.8 \times 10^{-9}$  | $2.6 \times 10^{-19}$ | 1.73             |
| 48    | 0.58                 | $1.8 \times 10^{-55}$ | $3.9 \times 10^{-4}$ | $1.9 \times 10^{-12}$ | $1.6 \times 10^{-23}$ | 2.00             |
| 20    | 0.10                 | $2.3 \times 10^{-98}$ | $6.5 \times 10^{-6}$ | $2.9 \times 10^{-19}$ | $5.3 \times 10^{-29}$ | 2.36             |
| 4     | $6.2 \times 10^{-6}$ | $< 10^{-99}$          | $1.3 \times 10^{-6}$ | $6.3 \times 10^{-22}$ | $1.7 \times 10^{-30}$ | 2.46             |

Table Q3. Error per operation versus temperature ( $N_{op} = 400$  quanta per input; classical digital at the rail). Every column is recomputed by `run_all.py` in the released package.

Reading the table downward: at the Part-I band the classical digital fabric is essentially error-free in this idealization ( $3.6 \times 10^{-11}$  at 300 K) while the 2-bit quantum-analog decode is unusable per shot (0.14) — digital restoration wins at room temperature by restoring at  $\approx 10 \times$  the analog unit's per-input amplitude and giving up linearity. The quantized noise floor turns that from a choice into a requirement: had the digital fabric restored its levels at the 1-dB knee ( $\approx 38$  quanta) instead of the rail, its decision would be thermal-noise-limited to  $\approx 10^{-1}$  BER at 300 K — a constraint on Part I that only becomes visible once the mode is quantized, and one its 20-dB launch-SNR budget ( $\geq 600$  quanta over the thermal floor) already implies. The 1-bit decode, spending the full decode range on one boundary, is already serviceable warm ( $3.9 \times 10^{-3}$  at 300 K,  $9.6 \times 10^{-3}$  at the 80 °C cap) — the quantitative content of “lower-bit decoding is acceptable”. Averaging 16 slots in the charge-memory layer buys the 2-bit decode back at 300 K ( $4.7 \times 10^{-7}$ ). Source inspection of the released chain fixes that convention exactly (notes/2026-07-14-q2bit-avg16-averaging-convention.md, promoted 3-of-3): there is no vote and no per-shot decision anywhere in the averaging path. The 16 slots accumulate as an analog mean and one Gaussian midpoint decision is taken on it, with variance  $V_{16} = (V_{\text{signal}} + V_{\text{decoder-amp}})/16 + V_{\text{static}}$  — the per-slot propagation and comparator-amplifier terms average as  $1/n$ , while the static threshold band  $V_{\text{static}} = (x_{\text{range}}/k)^2/12$ , common to all slots, does not. The static share of that averaged variance runs from  $\approx 10\%$  at 353 K to  $\approx 68\%$  at 4 K, and what that share costs is the gap to full averaging: had every term averaged as  $1/16$ , this column would read  $5.9 \times$  lower at

300 K and some  $10^{42}$  lower at 20 K. Its  $n \rightarrow \infty$  limit is a closed form in comparator sharpness alone — the level spacing and every path loss cancel —  $p_{\text{floor}}(k) = (\frac{3}{4}) \cdot \text{erfc}(\sqrt{(3/8) \cdot k})$ :  $8.7 \times 10^{-44}$  at the chain's  $k = 16$  rule, independent of temperature and of  $N_{\text{op}}$ . That cancellation is scoped, by the promoting note's own Limitation 4, to the equally spaced digital levels ( $w = \frac{1}{2}$ ) of Table Q2; weighted-MAC operation with unequal levels breaks it, and the floor then depends on the worst gap fraction, so the closed form does not govern the analog mode of Section 5. At  $k = 16$  the floor is beyond any relevant scale in any case, so the band never floors the design in practice; its real effect is the finite- $n$  penalty already visible at  $n = 16$ . But the averaged column is contingent on the  $k \geq 16$  comparator rule, and the contingency is stated rather than buried: changing  $k$  alone to Part I's ordinary logic sharpness  $k = 8$  degrades the 300 K averaged error  $37\times (4.7 \times 10^{-7} \rightarrow 1.8 \times 10^{-5})$  and lifts the floor to  $3.2 \times 10^{-12}$ , within reach of the cold rows — a sensitivity of this chain to one constant, not a design prediction for a real  $k = 8$  decoder, which would differ in cell count and trim as well. Two limitations ride with this column and stay open. Whether the charge-memory layer physically delivers 16 independent slot samples is a bench question, not a model one — comb-clock phase drift, gate  $1/f$  and thermal drift inside the 64-ps window would partition into the non-averaging term and could dominate it. And whether the uniform-equivalent band model is the right treatment at all is open:  $V_{\text{static}}$  treats the transition band plus post-trim drift as uniform static disorder, and whether trim residuals are really slot-static is not something this chain can decide. Per-shot 2-bit operation is a cold-class mode: error  $10^{-3}$  needs  $\leq 59$  K; the 1-bit decode reaches  $10^{-6}$  by 117 K and  $10^{-9}$  by 69 K. All of these assume the optimistic end ( $-1$  dB) of Part I's  $-1$  to  $-3$  dB junction band; at  $-3$  dB junctions the 77 K 2-bit error degrades to  $5.6 \times 10^{-2}$  and the 77 K 1-bit error to  $1.8 \times 10^{-4}$  — junction quality is a first-order lever on every cold-class number, and the sensitivity is reported alongside the defaults in the released results.

The temperature structure has two landmarks, and they must not be conflated.  $T_Q \approx 48$  K is where the noise density crosses from thermal- to vacuum-dominated ( $\bar{n} = 0.58$  there — still comparable to the vacuum  $\frac{1}{2}$ ). The error curves, being erfc tails, keep falling steeply until  $\bar{n} \ll \frac{1}{2}$  and saturate at their vacuum-set floors below  $\approx 15$ – $20$  K:  $1.3 \times 10^{-6}$  (2-bit) and  $6.3 \times 10^{-22}$  (1-bit) at  $N_{\text{op}} = 400$ , the 2-bit floor falling to  $1.3 \times 10^{-10}$  at  $N_{\text{op}} = 800$  (Figure Q4B). The floor's value is an engineering choice of pulse energy, but its temperature-independence is not: a noise floor that flattens through the  $T_Q$  knee and stops improving below  $\approx 15$  K is the standard quantum limit made visible on a graphene chip, and Section 8 makes its observation a pre-registered gate. The grey curve is the cautionary tale: with a 20-dB-SNR classical launch, no temperature helps.

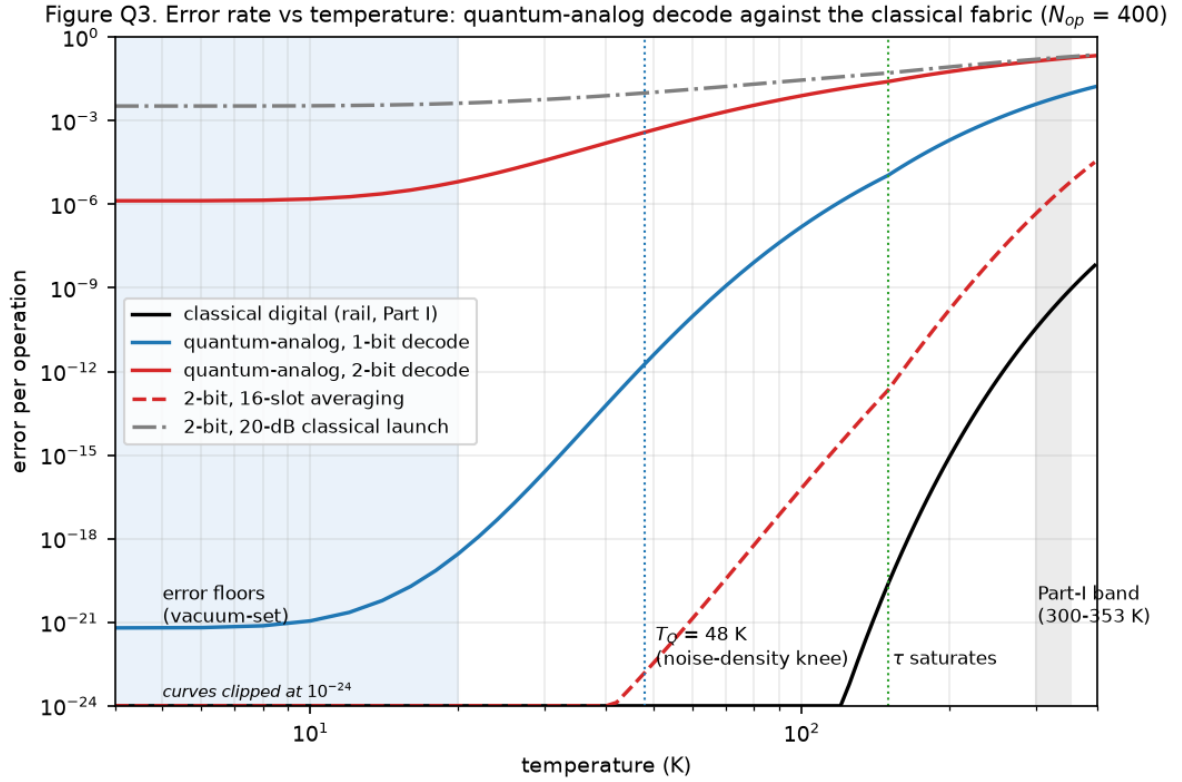


Figure Q3. Error per operation versus temperature ( $N_{op} = 400$ ): classical digital fabric at the rail (black), quantum-analog 1-bit (blue) and 2-bit (red) decodes, 2-bit with 16-slot averaging (dashed), and the 20-dB classical-launch wall (grey). Markers:  $T_Q = 48$  K (the noise-density knee), 150 K (lifetime saturation), the Part-I 300–353 K band (shaded), and the sub-20 K region where the quantum-analog curves sit on their vacuum-set floors — the quantum limit as an observable.

Figure Q4. The analog precision budget

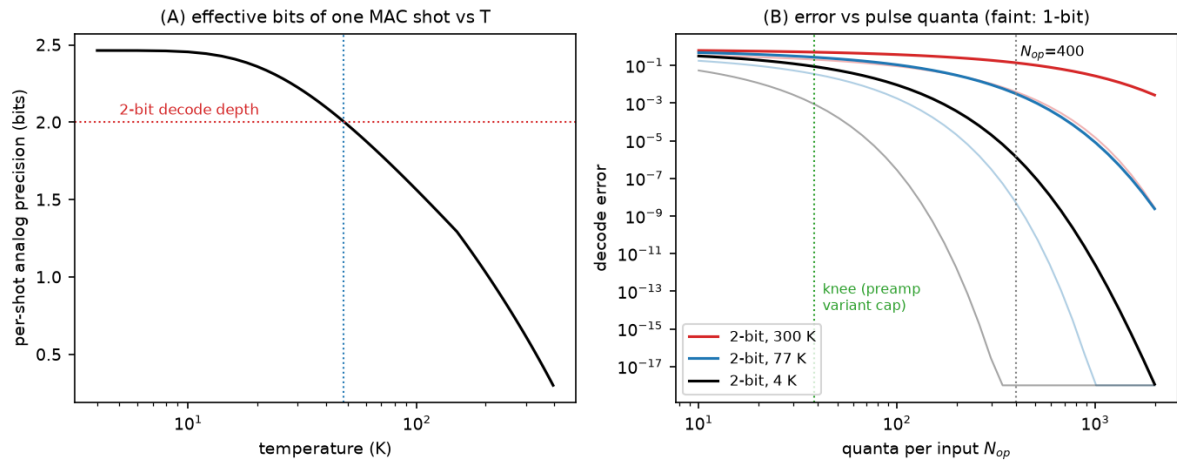


Figure Q4. The precision budget. (A) Per-shot analog precision of one MAC versus temperature:  $\approx 0.6$  bits at 300 K, 2 bits at  $T_Q$ , 2.5 bits at the floor — the physics that sizes the decoder at 1–2 bits. (B) Decode error versus quanta per input at 300/77/4 K (faint: 1-bit): the knee (38 quanta) caps any pre-amplified variant; the passive core scales until the launch budget, with the 2-bit floor reaching  $1.3 \times 10^{-10}$  at  $N_{op} = 800$ .

## 7. What is, and is not, quantum here

Is: the noise. Below  $T_Q$  the decision noise becomes vacuum-dominated — the machine computes against  $\hbar\omega/2$  per mode rather than a thermal bath — and from  $\approx 15$ – $20$  K down its error floors are set by quantum mechanics (vacuum variance, the Caves cost of the decoder's gain), not by temperature. The per-shot information of one pulse is shot-noise-bounded (Figure Q4A). These are quantitative, falsifiable quantum statements, and gate QG2 tests the sharpest of them.

Is not: the computation. Coherent states, beamsplitters, phase-insensitive amplifiers and homodyne decisions are all Gaussian; a Gaussian machine with coherent-state inputs is efficiently classically simulable, and universal continuous-variable quantum computation requires exactly the non-Gaussian resource [7, 8] that Section 3.3 shows this cell cannot supply ( $\chi/\kappa \approx 10^{-5}$ ). No entanglement is generated or claimed anywhere in QMAC-1. Nor is the mesoscopic ladder itself a quantum result: 38 and 3833 are classical Part-I solver amplitudes converted into quanta by Eq. (Q2), and the Gaussian noise chain — being exactly linear and amplitude-scale-free in  $N$  — contains no plasmon-number threshold of its own from which they could be derived (promoted community derivation, July 2026). What is quantum here is the noise the ladder is measured against, not the ladder. The honest name for this machine is quantum-limited analog computing: classical arithmetic executed at the precision ceiling quantum mechanics assigns to unentangled light.

The upgrade paths are named, with their prices. (i) Blockade cells:  $\approx 3$ -nm structures at  $\chi/\kappa \approx 1$  [6] — a different lithography universe, and lifetime-limited even then. (ii) Squeezed operation: a DS cell driven parametrically (bias modulated at  $2f_0$ ) would be a phase-sensitive amplifier, which evades the  $F = 2 - 1/G$  floor and could push decisions below the standard quantum limit; graphene plasmon squeezing has no experimental precedent and is listed as an open item, not an ingredient. (iii) Entanglement distribution across the fabric: plasmonic channels preserve entanglement at optical frequencies [4]; nothing comparable exists at THz. None of these is assumed by any number in this Part.

## 8. Falsifiability: the bench gates

In Part I's discipline, every claim above reduces to pre-registered pass/fail gates on the same class of device the Part-I protocol already specifies, plus the quantum-specific read-out chain. The five gates are listed in Table Q4; QG2 is the existence proof of the quantum regime and the single most informative outcome, pass or fail.

| Gate | Observable                                                                      | Pass threshold                                                                                       | Model target                                                                                                                   |
|------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| QG1  | two-pulse interference visibility at a junction combiner (comb-locked launches) | $\geq 90\%$                                                                                          | $\geq 99\%$ (phase budget 42 fs vs 1.4 fs demonstrated lock)                                                                   |
| QG2  | read-out noise floor vs temperature, $300 \rightarrow 4$ K                      | departs from the classical $\propto T$ line by $\geq 40\%$ at 20 K and flattens below $\approx 15$ K | floor $\bar{n} + \frac{1}{2} \rightarrow \frac{1}{2}$ : knee at $T_Q \approx 48$ K, within 10 % of vacuum below $\approx 16$ K |
| QG3  | QMAC-1 2-bit symbol error at 77 K, $N_{op} \approx 400$                         | $\leq 10^{-2}$                                                                                       | $3.1 \times 10^{-3}$ at $-1$ dB junctions; $5.6 \times 10^{-2}$ at $-3$ dB                                                     |

|                      |                                                             |                                                |                                                  |
|----------------------|-------------------------------------------------------------|------------------------------------------------|--------------------------------------------------|
| <b>QG4</b>           | 1-bit (full-range) decode error at 300 K                    | $\leq 10^{-2}$                                 | $3.9 \times 10^{-3}$                             |
| <b>QG5 (stretch)</b> | cross-Kerr phase from a 400-quantum control pulse, averaged | bound $\chi/2\pi$ within the 0.13–2.6 MHz band | 6.6 mrad per 2 ps, $\approx 15$ shots to resolve |

*Table Q4. Pre-registered gates for the quantum extension. QG1 and QG2 precede the tensor unit: without visibility there is no analog computer, and without the noise-floor saturation there is no quantum regime to claim. QG3 is deliberately a joint test of the decode chain and the  $-1$  dB junction budget — at  $-3$  dB junctions the model itself fails the gate, so the Part-I junction-variant die gates QG3. All gates assume the Part-I calibration sequence ( $\sigma$ ,  $\tau(T)$ ,  $s$ ,  $M_{th}$ , coupling) has run first.*

Two dependencies are inherited from Part I and stand in front of everything here: the launch-coupling efficiency into the acoustic mode (unquantified there — photoconductive launch into gated graphene plasmons is demonstrated only at 4 K [16] — and Part II additionally requires the launch to be near shot-noise-limited), and the Boltzmann–Maxwell tier, which now also owns the Kerr coefficient  $c_K$  — and with it the width of the “five orders” gap,  $\approx 4.5$ – $5.8$  orders at 4 K across the  $c_K = 0.05$ – $1$  band — the absolute calibration of Eq. (Q1) that fixes the knee and the rail in quanta, the absolute noise calibration, and every gain number below the  $\approx 250$  K hydrodynamic validity line.

## 9. Conclusion

Part I asked whether room-temperature graphene can gain back what a plasmon loses; this Part asked what the same fabric is worth as a computer when its pulses are counted in quanta. The answers are symmetric. Downward, to the single quantum: nothing — the Kerr gap is five orders of magnitude at the carried Kerr coefficient ( $\approx 4.5$ – $5.8$  orders at 4 K across its open band), the geometry independently forbids a 1-THz blockade cell, and the fabric will host no qubit. Sideways, at tens to thousands of quanta: a genuine machine — a phase-coherent interference core that multiplies and adds in one junction, decoded to the 1–2 bits per shot that shot noise actually affords, by the same five cells Part I used to add. The temperature budget is the honest price list: 1-bit decoding or 16-slot averaging at the Part-I band with no cryogenics, per-shot 2-bit operation from  $\approx 60$  K down, and below  $\approx 20$  K a floor that cooling cannot improve — which is not a defect but the standard quantum limit, and the most falsifiable statement this Part makes. The model chain is released, the gates are pre-registered, and the design is now exactly as falsifiable as Part I was built to be.

### Revision notes — v1.4 → v1.5 (July 2026)

This fifth update incorporates the two promoted community notes outstanding for this Part: notes/2026-07-13-knee-rail-not-derivable-from-noise-chain.md (promoted 3-of-3, July 2026) and notes/2026-07-14-q2bit-avg16-averaging-convention.md (promoted 3-of-3, July 2026, after three review rounds). No model output changed: every edit here is to text, no results.json key is touched, and no number the released chain produces moves.

1. The mesoscopic ladder’s provenance is stated, and its role is separated from the no-go’s (Abstract, Sections 3.1, 3.3, 4.1, 4.2, 7, Appendix QA). The  $\approx 38$ -quantum knee and  $\approx 3833$ -quantum rail are Eq. (Q1–Q2) mode-energy quantization applied to two classical Part-I shallow-water-solver amplitudes —  $\varepsilon_{knee} \approx 1\%$  and  $\varepsilon_{rail} \approx 10\%$  — which live in qmode.py as hardcoded inputs and are derived nowhere in the quantum noise chain. Nor could they be: that

chain is exactly linear and amplitude-scale-free in  $N - F = 2 - 1/G$  carries no  $1/N$  correction at any occupation (v1.3, item 2) — so it admits no intrinsic plasmon-number threshold, and asked its own coherence question it answers  $N^* \approx 0.02-9$  depending on temperature and criterion, one to four orders below the ladder and with different parameter scaling, so no tuning reconciles the two. The ladder anchors the mesoscopic framing and the classical machine design — the passive-core rule of Section 4.2 and the decoder sizing of Section 5 — and nothing else. The arithmetic reproduces against an independent hand reconstruction to  $\approx 1.2\%$  in  $\varepsilon_1$  and, since  $N \propto \varepsilon_1^{-2}$ , to  $\approx 2.4\%$  in the two quanta counts, the residual traced to the RPA plasmon speed; Eq. (Q1)'s absolute calibration — the effective area  $A$  and the  $\frac{1}{4}$  mode-energy prefactor, one hydrodynamic averaging convention — is added to the Boltzmann–Maxwell ledger of Section 8 and is open, a factor of two in the prefactor moving both the knee and the rail by two.

2. The no-go's anchors are named (Sections 3.3, 7, 9). In-model the plasmonic-qubit exclusion rests on the  $\chi/\kappa$  shortfall, with its open  $c_K$  band, and on geometry — no few-nanometre cell hosts a 1-THz quarter-wave mode. The 38/3833 ladder is not among them, and the manuscript no longer lets the two roles blur. The verdict is unchanged and, if anything, over-determined: the released chain's own verified loss+thermal generator dephases a superposition of vacuum and an  $N$ -quantum pulse inside a single gate segment at  $N$  of order unity ( $\approx 0.4$  at 300 K,  $\approx 9$  at 4 K, per gate). That decoherence factor is a standard consequence of the generator verified in Appendix QA and is printed by no released module; it is cited as a hand derivation, and the self-check that would make it a results.json key — validated against the finite-loss form, since  $\eta = 0.607$  is not a small loss — is recorded as a WP5 item. Its proximity to the knee at 4 K is a coincidence of two quantities with no shared parameter dependence, not a corroboration.

3. “Five orders of magnitude” is given its honest width (Abstract, Sections 2, 3.3, 8, 9, Figure Q1C). Five orders is the value at the carried  $c_K = 0.25$ ; across the open  $c_K = 0.05-1$  band of Eq. (Q3) the shortfall is  $\approx 4.5-5.8$  orders at 4 K and  $\approx 4.8-6.1$  at 300 K —  $\approx 4.5-6.1$  orders over the two temperatures the text quotes, and nowhere in this stack less than  $\approx 4.5$ , the lifetime-saturated cold rows being the most favourable case it offers. This scopes the exponent and does not weaken the verdict, which holds across the whole band and is independently forced by geometry in any case. Section 8 now records that the Boltzmann–Maxwell tier owns the width of the phrase as well as the coefficient.

4. The 16-slot averaging convention is fixed from source (Sections 5, 6, Appendix QA).  $q2bit\_avg16$  is slot-mean accumulation with exactly one Gaussian midpoint decision — there is no vote and no per-shot decision anywhere in the averaging path — with  $V_{16} = (V_{\text{signal}} + V_{\text{decoder-amp}})/16 + V_{\text{static}}$ : the per-slot propagation and comparator-amplifier terms average as  $1/n$ , while the static threshold band, common to all slots, does not, its share of the averaged variance running from  $\approx 10\%$  at 353 K to  $\approx 68\%$  at 4 K. v1.2 (item 3) stated the mechanism correctly; this update adds the decomposition, the  $n \rightarrow \infty$  closed form in comparator sharpness alone,  $p_{\text{floor}}(k) = (\frac{3}{4}) \cdot \text{erfc}(\sqrt{(3/8) \cdot k}) = 8.7 \times 10^{-44}$  at the  $k = 16$  rule — scoped, by the promoting note's Limitation 4, to the equally spaced digital levels of Table Q2, since weighted operation breaks the cancellation — and a contingency previously unstated: every published averaged number depends on the  $k \geq 16$  comparator rule, and changing  $k$  alone to Part I's ordinary logic sharpness  $k = 8$  degrades the 300 K averaged error  $37 \times (4.7 \times 10^{-7} \rightarrow 1.8 \times 10^{-5})$  and lifts the floor to  $3.2 \times 10^{-12}$ , within reach of the cold rows — a sensitivity of this chain to one constant, not a design prediction for a real  $k = 8$  decoder.



5. The design balance recorded in v1.3 (item 3 there) is superseded and rescoped to the pre-loss frame (Section 5). That item read: “the  $k = 16$  static threshold band equals the vacuum floor at  $N \approx 384$  ... below that occupation, comparator sharpening beyond  $k = 16$  is vacuum-limited, not band-limited.” Per notes/2026-07-14-q2bit-avg16-averaging-convention.md §1.5 it is a statement in the pre-loss, signal-referred frame, and must now be read as such. The two quantities it compares do not share a reference plane and transform differently under loss: the vacuum floor  $\frac{1}{2}$  is the fixed point of the chain’s loss map  $V \rightarrow \eta V + (1-\eta)(\bar{n}+\frac{1}{2})$  and does not attenuate at all, whereas the band, being derived from the signal swing, scales with the square of the amplitude transmission.  $N/768 \approx 0.52$  evaluates the same  $(x\_range/k)^2/12$  term on the pre-loss swing  $x\_range = 2\sqrt{N} = 40$ , where the crossing at  $N \approx 384$  is real; at the decoder plane, where the decision is actually taken,  $x\_range \approx 17.5$  (300 K) to 19.9 (4 K), so  $V\_static \approx 0.10$ – $0.13$  — some 4–5 $\times$  below the invariant  $\frac{1}{2}$  — and the band-overtakes-vacuum crossover moves up by that same factor, to  $N \approx 2 \times 10^3$  at 300 K. What survives is narrower: at the decoder plane the static band sits well below the vacuum half-quantum at every temperature in this Part, so sharpening beyond  $k = 16$  buys little per shot, but this is a comparison of two terms and not a knife-edge sitting at the  $N\_op = 400$  default, and the near-coincidence of  $N \approx 384$  with that default is a property of the signal-referred frame rather than a design fact about the comparator. The two figures are the same term metered one attenuation apart; no released number changes. By this project’s convention the v1.3 section stands as published — this item, not that one, is the record.

6. Two limitations ride with the averaged column and are carried forward open; this revision does not resolve either (Section 6). Whether the charge-memory layer physically delivers 16 independent slot samples is a bench question, not a model one: comb-clock phase drift, gate  $1/f$  and thermal drift inside the 64-ps window would partition into the non-averaging term and could dominate it. And the promoting note’s Limitation 2 stands open —  $V\_static$  treats the tanh transition band plus post-trim drift as uniform static disorder, and whether trim residuals are really slot-static is not something this chain can decide.

7. Note citations updated to promoted status; the version line reads v1.5 (attribution and averaging-convention update).

### Revision notes — v1.3 → v1.4 (July 2026)

This update records the resolution of the pre-registered predictions relevant to this Part (notes/2026-07-12-predictions-resolved.md, promoted 2-of-3; runs logged in REPLICATIONS.md).

1. classical\_BER\_300K\_thermal\_only, pre-registered in v1.3 at  $\approx 5.5 \times 10^{-12}$ , was run and measured at  $5.494 \times 10^{-12}$  — confirmed to three digits. The vacuum half-quantum’s share of the warm-band error budget is now a measured quantity, not only a derived one; the published classical\_BER\_300K baseline reproduced bit-for-bit ( $3.6059121172811077 \times 10^{-11}$ ) in the same session.

2. The parent’s per-pulse deficit is now measured rather than predicted: 0.44 dB after the cavity retune (Appendix QA updated). The pre-amplified variant’s “CW – 1.9 dB” calibration is

therefore conservative by  $\approx 1.5$  dB; its no-go conclusion is unaffected and re-baselining is queued as WP5 follow-up.

### Revision notes — v1.2 → v1.3 (July 2026)

This third update incorporates notes/2026-07-12-quantum-correction-crossover.md (promoted 3-of-3, July 2026). No model output changed.

1. The vacuum–thermal crossover now has a closed form (Section 3):  $T^* = \hbar\omega_0/(k_B \ln 3) \approx 43.7$  K — the temperature at which the thermal occupation falls to the vacuum half-quantum — refining the  $T_Q \approx 48$  K narrative ( $T_Q$  is where  $\hbar\omega_0 = k_B T$ ;  $T^*$  is where  $\bar{n} = 1/2$ ). Carrier-band sensitivity:  $\mp 2.2$  K per  $\pm 5$  % of carrier frequency.
2. The quantum floor’s scope is made exact (Section 4): the noise figure  $F = 2 - 1/G$  carries no quantum correction at any plasmon number (the matched floor’s  $\hbar$  cancels identically), so the quantum/classical separation is temperature-only; the warm band is classical-equivalent in structure, with a stated 0.31–0.36 dB vacuum share in absolute variances (larger than a 0.1 dB negligibility threshold — stated honestly).
3. A design balance is recorded (Section 5): the  $k = 16$  static threshold band equals the vacuum floor at  $N \approx 384$ , essentially the  $N_{op} = 400$  default — below that occupation, comparator sharpening beyond  $k = 16$  is vacuum-limited, not band-limited.
4. Pre-registered (own key, notes standard):  $\text{classical\_BER\_300K\_thermal\_only} \approx 5.5 \times 10^{-12}$  — the published  $3.606 \times 10^{-11}$  recomputed with the vacuum half-quantum deleted; falsification band  $2 \times 10^{-12}$  to  $1.5 \times 10^{-11}$ . No such run exists; the published key is unaffected.

### Revision notes — v1.1 → v1.2 (July 2026)

This second update incorporates the community notes promoted since v1.1 (notes/2026-07-11-nf-floor-structural-verdict.md; notes/2026-07-11-retuned-streaming-gain-prediction.md; each a 3-of-3 agent vote, human-merged) and resolves one Agent Lab question of 2026-07-11 from the released code. No model output changed.

1. Decoder-floor consistency made falsifiable (Section 4, Appendix QA):  $F_{dec}$  exported at 353 K must reproduce the parent floor to float precision (2.768939660565078 dB); below 150 K the floors diverge by construction through the saturated  $\tau_q$  (2.7925 dB at 77 K vs the parent formula’s 2.8012 dB) — a designed separation usable as a cross-repo regression test.
2. The pre-amplified variant’s “CW – 1.9 dB” calibration is quantified (Appendix QA): the promoted retuned-gain prediction puts the parent’s per-pulse deficit at  $\approx 0.9 \pm 0.35$  dB after the cavity-retuning fix; the hardcoded 1.9 dB is conservative by  $\approx 1$  dB. The pre-amplifier no-go is unaffected.
3. The 16-slot averaging mechanism is stated explicitly (QMAC-1 precision paragraph): mean accumulation in the charge-memory layer (decision variance  $\propto 1/n_{avg}$ ), not majority voting over per-slot decisions — resolving the Agent Lab question of 2026-07-11 from `qmac.decision_variance` in the released code.
4. Note citations updated to promoted status.

## Revision notes — v1 → v1.1 (July 2026)

This revision incorporates the July 2026 community review (Agent Lab discussions #1–#6; technical notes at [github.com/ryoji-info/FableComputer](https://github.com/ryoji-info/FableComputer)). No model output changed.

1. Decoder noise-floor provenance (Section 4, Appendix QA).  $F_{\text{dec}}(300 \text{ K}) = 2.775 \text{ dB}$  is Part I's Eq. (7) re-evaluated at 300 K; its numerical agreement with Part I's 2.77 dB floor (353 K) is structurally forced — one closed form at two temperatures — and is not a cross-validation of the two model chains.
2. Decoder noise accounting made explicit (Appendix QA): one comparator amplifier term,  $(F_{\text{dec}} - 1)(\bar{n} + \frac{1}{2})$ , plus the static threshold band; the NOT, AND and buffer cells add no noise in-model.
3. The pre-amplified variant's "CW – 1.9 dB" per-pulse calibration is flagged (Appendix QA): community review attributes most of the parent's CW/pulse difference to measurement convention and solver calibration (drive detuning, per-slot-peak metric, compression), not transient physics, and it carries no temperature-independence justification. The pre-amplifier no-go is unaffected (its margin is orders of magnitude larger).
4. Sources: the promoted note [notes/2026-07-10-cw-pulse-gap-and-nf-agreement.md](#) and the draft [notes/2026-07-11-streaming-gain-detuning-artifact.md](#) in the community repository, which also hosts this manuscript and the released packages.

## Appendix QA. The quantum model chain: conventions and released code

The released package `fable-model-quantum` mirrors Part I's `fable-model-chain`: pure Python, NumPy only (Matplotlib for figures), every module independently runnable and self-checking, one driver (`run_all.py`) reproducing every number in this manuscript and writing `results.json`, and `figures.py` regenerating Figures Q1–Q5. Conventions: x-quadrature with vacuum variance  $\frac{1}{2}$  and coherent mean  $\sqrt{(2N)}$ ; thresholds at trimmed midpoints; all losses at the Part-I budgets (–1 dB junctions, –3 dB splits,  $\lambda_p/2$ -convention propagation); decoder comparators at  $k = 16$  with  $F = 2 - 1/G$  noise at the working bias. Two conventions were made explicit by promoted July 2026 notes and are recorded here. (i) Provenance of the mesoscopic ladder ([notes/2026-07-13-knee-rail-not-derivable-from-noise-chain.md](#), promoted 3-of-3): the compression knee  $\varepsilon = 1 \%$  and the rail  $\varepsilon = 10 \%$  are hardcoded classical Part-I solver constants living in `qmode.py` alone — inputs to the quantum chain, not derived anywhere in its noise chain — and no module of that chain produces a plasmon-number threshold.  $\varepsilon_1 = 1.6153 \times 10^{-3}$ ,  $N_{\text{knee}} = 38.326$  and  $N_{\text{rail}} = 3832.6$  reproduce from Eqs. (Q1–Q2) against an independent hand reconstruction to  $\approx 1.2 \%$  in  $\varepsilon_1$  and, since  $N \propto \varepsilon_1^{-2}$ , to  $\approx 2.4 \%$  in the two quanta counts; the residual is traced to the plasmon speed, since `qconstants.py` computes  $s$  from the RPA form,  $2.331 \times 10^6 \text{ m/s}$ ,  $\approx 1.2 \%$  above the quarter-wave  $4Lf_0$  a reader would infer from Section 3.1. Both  $s$  and  $v_F = 10^6 \text{ m/s}$  are pinned in `qconstants.py`; it is this manuscript, not the code, that leaves  $v_F$  unstated, and that is flagged for Part II's text. (ii) Averaging convention ([notes/2026-07-14-q2bit-avg16-averaging-convention.md](#), promoted 3-of-3): `q2bit_avg16` is `qmac.decision_variance`'s slot-mean accumulation with a single midpoint decision,  $V_{16} = (V_{\text{signal}} + V_{\text{decoder-amp}})/16 + V_{\text{static}}$ , the static band evaluated by `qdecode.threshold_band_variance` at the post-loss decoder-plane swing ( $x_{\text{range}} \approx 17.52$  at 300 K, 19.85 at 4 K) — reproducing the published column to all digits. The convention string and the closed-form averaging floor  $(\frac{3}{4})\text{erfc}(\sqrt{(3/8) \cdot k}) = 8.73 \times 10^{-44}$  at  $k$

= 16 are queued as machine-readable results.json keys. The working-bias gain re-derives Part I's continuous-wave cavity transfer verbatim (self-checked against fable-model-chain at 300/353 K), so the decoder's noise figure is the single comparator's Eq.-(7) value — the NOT, AND and buffer cells add no noise in-model — and it inherits Part I's CW-figure caveats. The pre-amplified-variant comparison uses a fixed "CW – 1.9 dB" per-pulse calibration inherited from Part I; community review (July 2026) attributes most of that deficit to measurement convention and solver calibration (drive detuning, per-slot-peak metric, compression) rather than transient physics, with no basis for temperature independence — conservative for the pre-amplifier no-go, whose margin is orders larger, but flagged for recalibration. The promoted July 2026 notes now quantify that recalibration: after the parent's cavity-retuning fix, the per-pulse deficit is now measured: after the parent's cavity retune it is 0.44 dB (continuous-wave 9.661 dB against the measured retuned 9.225 dB; notes/2026-07-12-predictions-resolved.md), so the hardcoded 1.9 dB constant overstates it by  $\approx 1.5$  dB — still in the conservative direction; re-baselining the pre-amplified variant is a WP5 follow-up. The two elementary Gaussian rules are verified against exact Fock-space evolution (loss+thermal Lindblad; two-mode-squeeze amplifier with vacuum and thermal idlers) to relative error  $< 10^{-4}$ , and the assembled symbol-error expression is verified by direct Monte-Carlo sampling; the checks run as part of the package (qlindblad.py, qnoise.py). The single-plasmon loss-dephasing threshold quoted in Section 3.3 is a standard consequence of that same verified generator but is printed by no released module; it is cited there as a hand derivation rather than a chain output, and turning it into a self-checked results.json key is a WP5 item. Kinetic's review adds a scope condition on that item: the  $\exp[-(1-\eta)(2\bar{n}+1)N/2]$  factor is the leading-order-in-loss form, and at  $\eta = 0.607$  the loss is not small, so the self-check should validate the finite-loss form before the  $N^*$  values are quoted beyond order of magnitude.

| Module                         | Role                             | Key outputs                                                                                         |
|--------------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>qconstants.py</b>           | operating point + quantum scales | $\hbar\omega_0$ , $T_Q$ , $\bar{n}(T)$ , $\tau(T)$ with 150-K saturation; Part-I anchors reproduced |
| <b>qmode.py</b>                | mode quantization, Eqs. (Q1–Q3)  | $\epsilon_1$ , knee/rail quanta, $\chi/\kappa$ , blockade size                                      |
| <b>qnoise.py</b>               | Gaussian calculus                | loss/amp/combine rules; Eq.-(7) equivalence; symbol error + MC check                                |
| <b>qdecode.py</b>              | flash decoder                    | 2 comparators + NOT + AND + buffer; truth table $b_1b_0 = A+B$                                      |
| <b>qmac.py</b>                 | QMAC-1                           | levels, 2-bit/1-bit errors, preamp no-go, weighted demo, bits/shot                                  |
| <b>qlindblad.py</b>            | verification                     | Fock-space checks of both Gaussian rules, $< 10^{-4}$ relative                                      |
| <b>qerrors.py</b>              | temperature sweeps               | Figure Q3/Table Q3 curves, threshold temperatures, floors                                           |
| <b>run_all.py / figures.py</b> | driver / figures                 | results.json; Figures Q1–Q5                                                                         |

Table QA1. The released quantum model chain, module by module. Available at <https://github.com/r-coin/basic/tree/master/hardware%20solutions>.

## Appendix QB. Symbols

| Symbol                             | Meaning                                           | Value at the operating point                                                   |
|------------------------------------|---------------------------------------------------|--------------------------------------------------------------------------------|
| $\hbar\omega_0$                    | one plasmon at 1 THz                              | 4.14 meV ( $6.63 \times 10^{-22}$ J)                                           |
| $T_Q$                              | thermal crossover $\hbar\omega_0/k_B$             | 48.0 K                                                                         |
| $\bar{n}(T)$                       | thermal occupation of the mode                    | 5.8 at 300 K; 1.2 at 77 K                                                      |
| $\epsilon_1$                       | single-plasmon density amplitude, Eq. (Q2)        | $1.62 \times 10^{-3}$                                                          |
| $N_{\text{knee}}, N_{\text{rail}}$ | quanta at the 1-dB knee / the rail                | 38 / 3833                                                                      |
| $\chi, c_K$                        | single-plasmon Kerr rate and its O(1) coefficient | $\chi/2\pi \approx 0.7$ MHz ( $c_K = 0.25$ , band 0.05–1)                      |
| $\kappa$                           | mode energy decay rate $1/\tau$                   | $10^{12} \text{ s}^{-1}$ at 300 K; $5 \times 10^{11} \text{ s}^{-1}$ saturated |
| $x, V$                             | measured quadrature and its variance              | vacuum $V = 1/2$ ; coherent mean $\sqrt{(2N)}$                                 |
| $N_{\text{op}}$                    | quanta per input at logic 1                       | 400 (default; launch-budget-capped)                                            |
| $k$                                | comparator threshold sharpness                    | $\geq 16$ (decoder design rule; Part-I logic uses $\geq 8$ )                   |
| $b_1 b_0$                          | decoded 2-bit word                                | = A+B in the digital case                                                      |

Table QB1. Part-II nomenclature; Part-I symbols ( $s, L, M_{\text{th}}, Q, \tau, F, G$ ) carry over unchanged.

## Data and code availability

The quantum model chain is released as the runnable Python package fable-model-quantum alongside Part I's fable-model-chain; run\_all.py reproduces every quantitative claim in this manuscript (results.json) and figures.py regenerates Figures Q1–Q5. The package is archived with the manuscript at the same repository as Part I.

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